

Interpreting effective homogeneous rheologies through a local differential map with application to the Moon

Yeva Gevorgyan¹★

¹King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia

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ABSTRACT

Layered viscoelastic interiors control the frequency dependence of planetary tidal dissipation, but repeated layered Love-number calculations are expensive in long-term orbital and spin integrations. Equivalent homogeneous rheologies provide a cheaper representation of the same response, but their effective parameters do not by themselves identify the interior layers they represent. We develop a local interpretation of such parameters by differentiating the layered-to-homogeneous construction about a fixed-geometry baseline, perturbing only selected layer rigidities and viscosities. We use the pole-residue representation of the degree-two Love number to construct a physical-to-effective transfer matrix. We then use its singular values, resolution matrix, and mode-dominance matrix to determine which physical perturbations are visible in the Love-number response and how they appear in effective-rheology coordinates. Applied to a five-layer lunar model, the full degree-two response is sensitive to seven of eight solid-layer rheological parameters. The single near-null direction is the inner-core rigidity, consistent with shielding by the fluid outer core and with the weak dependence of Love numbers on core elastic structure. Over the LLR frequency interval, fewer physical directions are visible. To first order, the elastic spring, isolated dashpot, and two Voigt elements reproduce the nine-pole response over this interval. Finite perturbations confirm the tangent prediction to within a few percent, with the validity radius limited by a nearly degenerate pair of long-timescale modes.

Key words: planets and satellites: interiors – planets and satellites: dynamical evolution and stability – planets and satellites: individual: Moon – methods: analytical – methods: numerical

1 INTRODUCTION

Tidal dissipation is a primary driver of the long-term orbital and rotational evolution of planets and satellites, from the Solar System to close-in exoplanets (Bagheri et al. 2022; Boué & Efroimsky 2019; Correia et al. 2014). Its magnitude and variation with frequency depend on the deformable body’s internal structure and rheology. Real interiors are radially layered, so a layered model is the natural forward description for tidal deformation and dissipation (Sabadini et al. 2016; Henning & Hurford 2014). The drawback of layered models is computational, since each Love-number evaluation requires a full deformation solve (Tobie et al. 2005), repeated many times in a secular (long-term, orbit-averaged) or time-domain orbital integration. These integrations are costly, so dynamical studies often resort to highly simplified homogeneous models—a single Maxwell element or a constant quality factor—that are cheap to evaluate but cannot reproduce the frequency-dependent dissipation of a stratified interior, missing features such as the additional dissipation peaks contributed by individual layers (Efroimsky & Lainey 2007; Bolmont et al. 2020; Gevorgyan 2021).

This computational bottleneck can be reduced by replacing the layered body with a homogeneous body with a complex rheology. The interior is not physically homogeneous. Instead, an effective homogeneous model reproduces the same frequency-dependent Love

number. Layered bodies with finitely many Maxwell layers and a simple relaxation spectrum can be represented by equivalent generalized Maxwell or generalized Voigt models (Ragazzo & Ruiz 2017; Gevorgyan et al. 2023; Gevorgyan 2026c). We use finite-mode approximations to extend this representation to continuous-spectrum rheologies such as Andrade and Sundberg–Cooper (Gevorgyan et al. 2020; Gevorgyan & Ragazzo 2026).

For each interior, we construct an effective rheology whose parameters reproduce the Love number but carry no physical interpretation on their own. An effective viscosity or rigidity may reflect a single interior layer, several layers, or no layer in particular, and they are not simply layer averages—averaging the layer rigidity and viscosity over- or under-estimates the dissipation depending on structure, and reproducing the layered response requires fitted values instead (Bolmont et al. 2020; Henning & Hurford 2014).

We differentiate the pointwise reduction from a layered Maxwell body to its effective homogeneous generalized Voigt model about a chosen baseline to determine how the effective parameters change under perturbations of the physical layer rheologies. We use the resulting transfer matrix to measure how strongly each effective parameter is tied to a physical layer and which interior perturbations strongly or weakly affect the Love-number response. The geometry and density structure are held fixed and only the layer rigidities and viscosities are perturbed, which isolates the rheological differential map and matches applications where a reference geometry is chosen first and the uncertain quantities are the layer rheologies. We use

★ E-mail: yeva.gevorgyan@kaust.edu.sa

the same differential to propagate this uncertainty to the effective parameters used downstream.

Implicit differentiation of simple roots, eigenvalue sensitivity, and resolution diagnostics from linear inverse theory are classical tools. We combine them with the homogeneous-rheology inversion and express the resulting derivative in effective-rheology coordinates rather than directly in observable space. This distinguishes the present construction from standard sensitivity kernels of tidal response to interior structure, including direct Love-number partial derivatives (Okubo & Saito 1983; Tobie et al. 2005; Lau & Al-Attar 2021) and relaxation-mode kernels in glacial isostatic adjustment (Peltier 1976; Tromp & Mitrovica 1999; Al-Attar & Tromp 2014).

In Section 2, we recall the pole–residue representation of the layered Love number and the algebraic generalized Voigt inversion. We derive the layered-model spectral sensitivities in Section 3 and differentiate the homogeneous inversion to obtain the local transfer matrix in Section 4. We give a minimal two-layer example in Section 5, apply the method to the Moon in Section 6, and discuss interpretation, limitations, and the additional requirements for observational inference in Section 7.

2 THE POINTWISE REDUCTION AND THE LOCAL MAP

The reduction consists of two stages (Gevorgyan et al. 2023; Gevorgyan 2026c,a). In the first, the pole–residue spectral state u is obtained from the physical layer rheologies v through the Love number $k_2(s)$ (Section 2.1). The effective parameters w are then recovered from u by the algebraic generalized Voigt inversion $\mathcal{I}$ (Section 2.2). The local transfer matrix is the derivative of the composed map $v \mapsto u(v) \mapsto w(v)$ (Section 2.3).

2.1 Layered tidal response

Consider a spherically symmetric incompressible body made of finitely many homogeneous layers, with fixed radii and densities. Solid layers are modeled with Maxwell rheology. For liquid layers, the same deformation equations are evaluated in the zero-shear limit (Jara-Oru  & Vermeersen 2011). The degree-two tidal Love number is obtained from the standard propagator-matrix solution of the linear deformation equations (Sabadini et al. 2016). Under the usual simple-spectrum assumptions, this gives a rational function of the Laplace variable with the partial-fraction expansion (Gevorgyan et al. 2023, Eq. 26)

$$k_2(s) = k_E + \sum_{i=1}^{n+1} \frac{r_i}{s - s_i}, \quad (1)$$

where $0 > s_1 > s_2 > \dots > s_{n+1}$ are the relaxation poles and $r_i > 0$ are the corresponding residues. The elastic and fluid Love numbers are the high- and zero-frequency limits,

$$k_E = \lim_{|s| \rightarrow \infty} k_2(s), \quad k_f = k_2(0) = k_E - \sum_{i=1}^{n+1} \frac{r_i}{s_i}. \quad (2)$$

The imaginary part of $k_2(i\omega)$ gives the dissipative part of the tidal response at forcing frequency ω .

2.2 Equivalent generalized Voigt model

The equivalent homogeneous body has a generalized Voigt rheology, following the layered-to-homogeneous construction of (Ragazzo & Ruiz 2017; Gevorgyan et al. 2023; Gevorgyan 2026c). It consists of an

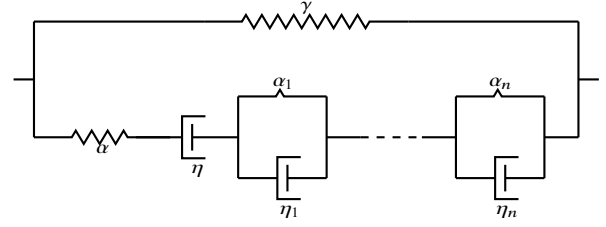


Figure 1. Schematic representation of the generalized Voigt model used as the equivalent homogeneous body.

elastic spring α , an isolated dashpot η , and n Kelvin–Voigt elements (spring α_i , dashpot η_i) in series, all in parallel with a gravitational spring γ (Figure 1). The number of these elements is fixed by the pole structure of the layered response, so the resulting rheological model has the same $n + 1$ relaxation poles as (1).

The self-gravitating body compliance $C(s)$ satisfies

$$k_2(s) = \Theta C(s), \quad C(s) = \frac{1}{\alpha + \gamma} + C_v(s), \quad (3)$$

where $\Theta = 3I_0 G/R^5$ is fixed by the body’s geometry and density, $1/(\alpha + \gamma)$ is the instantaneous compliance, and $C_v(s)$ is the frequency-dependent viscoelastic part, with $C_v(s) \rightarrow 0$ as $|s| \rightarrow \infty$ (Ragazzo 2020).

The gravitational and elastic rigidities are determined by the zero- and high-frequency limits of $C(s)$ (Ragazzo & Ruiz 2017; Ragazzo 2020),

$$\gamma = \frac{\Theta}{k_f}, \quad \alpha = \frac{\Theta}{k_E} - \gamma. \quad (4)$$

Since Θ and k_f are fixed, γ is fixed as well.

To extract the remaining viscoelastic parameters η and $\{\alpha_i, \eta_i\}_{i=1}^n$, we use the complex compliance of the generalized Voigt rheology,

$$\widehat{J}(s) = \frac{1}{\alpha} + \frac{1}{\eta s} + \sum_{i=1}^n \frac{1}{\alpha_i + s\eta_i}. \quad (5)$$

The generalized Voigt rheology is connected in parallel with the gravitational spring. The two compliances are therefore related by

$$C(s) = \frac{1}{\gamma + 1/\widehat{J}(s)}. \quad (6)$$

With $k_2 = N/P$ from (1), where P and N have degree $n + 1$ and P is monic, the parallel-spring identity gives

$$\widehat{J}(s) = \frac{k_2(s)}{\Theta - \gamma k_2(s)} = \frac{N(s)}{\Theta P(s) - \gamma N(s)}. \quad (7)$$

With $\gamma = \Theta/k_f$, the denominator in (7) has a simple zero at $s = 0$. We write

$$\Theta P(s) - \gamma N(s) = s Q(s), \quad (8)$$

where Q has degree n and negative simple roots $q_1, \dots, q_n$. The polynomial Q is used with this normalization, not as a monic polynomial. Its leading coefficient is $\Theta - \gamma k_E = k_E \alpha$. Residue matching in (7) gives

$$\eta = \frac{Q(0)}{N(0)}, \quad \eta_i = \frac{q_i Q'(q_i)}{N(q_i)}, \quad \alpha_i = -q_i \eta_i. \quad (9)$$

Equations (4) and (9) define the pointwise inversion map $\mathcal{I}$ from the spectral data (k_E, s_i, r_i) to the generalized Voigt parameters. We differentiate this map in Section 4.

2.3 The transfer matrix

Let $v \in \mathbb{R}^p$ be the physical parameter vector formed by the viscosities and rigidities of the solid Maxwell layers. With ℓ solid layers, we take

$$v = (\eta^{(1)}, \mu^{(1)}, \dots, \eta^{(\ell)}, \mu^{(\ell)})^T, \quad p = 2\ell, \quad (10)$$

where parenthesized superscripts index physical layers and distinguish the layer parameters $\mu^{(k)}, \eta^{(k)}$ from the effective parameters α_i, η_i . Liquid layers are part of the fixed reference model and are not perturbed.

From v , we compute the spectral state through the Love number $k_2(s; v)$,

$$u = (k_E, s_1, r_1, \dots, s_{n+1}, r_{n+1})^T, \quad (11)$$

and then apply the inversion $\mathcal{I}$ to obtain

$$w = (\gamma, \alpha, \eta, \alpha_1, \eta_1, \dots, \alpha_n, \eta_n)^T. \quad (12)$$

We linearize the composition $v \mapsto u(v) \mapsto w(v)$ about a baseline model v_0 , with induced spectral state $u_0 = u(v_0)$ and effective parameters $w_0 = \mathcal{I}(u_0)$.

We assume that the simple-spectrum assumptions of Section 2.1 hold at v_0 . This is the generic case for finite incompressible multilayer Maxwell bodies under the usual tidal-stability conditions, excluding density inversions between adjacent layers that can trigger positive roots (Sabadini et al. 2016, Sec. 1.8). Related stability analyses are given by Wu & Peltier (1982) and Tanaka et al. (2006). By the interlacing property of Gevorgyan et al. (2023, Eq. 22), the roots of Q are simple. With positive residues and fixed pole and element ordering, no effective element merges with another or carries zero weight. Under these assumptions, the forward spectral map $w \mapsto u$ is smooth with nonsingular Jacobian $J_w = \partial u / \partial w$, so its inverse $\mathcal{I} : u \mapsto w$ is a local diffeomorphism at u_0 .

Throughout the paper, $Df(x_0)$ denotes the Fréchet derivative of a smooth map f at x_0 , i.e., the Jacobian matrix $\partial f / \partial x(x_0)$. The derivative of the local reduction is written

$$\Delta w \approx T \Delta v. \quad (13)$$

Since $w = \mathcal{I}(u)$, the chain rule gives

$$T = D\mathcal{I}(u_0) Du(v_0). \quad (14)$$

Equivalently, differentiating the local inverse relation between the generalized Voigt parameters and the spectral data gives

$$T = J_w^{-1} J_v, \quad J_w = \frac{\partial u}{\partial w}(w_0), \quad J_v = \frac{\partial u}{\partial v}(v_0). \quad (15)$$

The algebraic inversion uses the complete coordinate sets $(k_E, s_i, r_i) \mapsto (\gamma, \alpha, \eta, \alpha_i, \eta_i)$, which have the same dimension. For fixed geometry and density, $Dk_f = 0$, hence $D\gamma = 0$, and one may drop the γ row and recover k_E from the pole-residue variables using the fixed value of k_f in (2). On the resulting $2n + 2$ coordinates, J_w is square and nonsingular, so the pseudoinverse J_w^+ used in the implementation coincides with J_w^{-1} .

3 SPECTRAL SENSITIVITIES OF THE LAYERED MODEL

The spectral Jacobian $J_v = Du(v_0)$ has rows given by the Fréchet derivatives of the spectral coordinates k_E, s_i , and r_i with respect to the physical layer parameters. Standard sensitivity analyses often differentiate the Love number itself, giving direct or adjoint kernels for $Dk_2(s; v_0)$ (Okubo & Saito 1983; Lau & Al-Attar 2021). Here we instead differentiate the pole-residue coordinates of k_2 , which

gives the mode-resolved input to the effective-rheology inversion of Section 4.

For a finite multilayer Maxwell body, the propagator-matrix construction gives a rational Love number $k_2(s, v) = N(s, v)/P(s, v)$, where N and P are polynomials in s (Sabadini et al. 2016; Gevorgyan et al. 2023). The $n + 1$ roots of P are the relaxation poles. In monic form,

$$P(s, v) = s^{n+1} + b_n(v)s^n + b_{n-1}(v)s^{n-1} + \dots + b_1(v)s + b_0(v), \quad (16)$$

so that $P(s, v_0) = \prod_{i=1}^{n+1} (s - s_i)$ at the baseline.

The physical rheological parameters enter through the complex shear moduli of the solid layers,

$$\hat{\mu}_k(s) = \frac{\mu^{(k)} \eta^{(k)} s}{\mu^{(k)} + \eta^{(k)} s}, \quad (17)$$

with

$$\frac{\partial \hat{\mu}_k}{\partial \mu^{(k)}} = \frac{(\eta^{(k)})^2 s^2}{(\mu^{(k)} + \eta^{(k)} s)^2}, \quad \frac{\partial \hat{\mu}_k}{\partial \eta^{(k)}} = \frac{(\mu^{(k)})^2 s}{(\mu^{(k)} + \eta^{(k)} s)^2}. \quad (18)$$

Let $F(s, \hat{\mu}_1, \dots, \hat{\mu}_\ell)$ denote the characteristic determinant of the propagator-matrix solution. At fixed s , its derivative with respect to a physical parameter is

$$\left. \frac{\partial F}{\partial v_j} \right|_s = \sum_k \frac{\partial F}{\partial \hat{\mu}_k} \frac{\partial \hat{\mu}_k}{\partial v_j}. \quad (19)$$

The derivatives $\partial F / \partial \hat{\mu}_k$ are obtained from the propagator-matrix construction, either symbolically or by finite differences at the baseline. After clearing the Maxwell denominators, removing common factors, and normalizing P to be monic, this gives the derivatives P_{v_j} used below. The same procedure applied to the numerator determinant gives N_{v_j} .

We first compute the pole sensitivities. Each pole is a simple root of the characteristic equation, equivalently $P(s_i(v), v) = 0$. Implicit differentiation gives

$$\frac{\partial s_i}{\partial v_j} = - \frac{P_{v_j}(s_i, v_0)}{P_s(s_i, v_0)}. \quad (20)$$

This quotient is invariant under multiplication of P by any factor that is nonzero at the pole, including factors that depend on s or v , because $P = 0$ at the pole. It may therefore be evaluated either in the monic normalization or directly from the cleared characteristic determinant, provided no pole coincides with a cleared Maxwell factor and the common-factor structure is locally unchanged.

Equivalently, (20) is the simple-eigenvalue sensitivity formula for the companion matrix of (16) (Lancaster 1964; Frank 1978; Nelson 1976). If poles cluster or coincide, the scalar quotient breaks down and one must differentiate the corresponding invariant subspace instead (Nelson 1976). This degenerate case is excluded by the simple-spectrum assumptions.

For a simple pole $s_i(v)$ of $k_2 = N/P$, the corresponding residue is

$$r_i(v) = \frac{N(s_i(v), v)}{P_s(s_i(v), v)}. \quad (21)$$

Differentiating this expression gives

$$\frac{\partial r_i}{\partial v_j} = \frac{N_{v_j} + N_s \partial_{v_j} s_i}{P_s} - \frac{N (P_{sv_j} + P_{ss} \partial_{v_j} s_i)}{P_s^2}. \quad (22)$$

For the Jacobian J_v , this expression is then evaluated at $v = v_0$ and $s_i = s_i(v_0)$.

The elastic coordinate k_E is the high-frequency limit of $k_2(s, v)$. Since $\widehat{\mu}_k(s)$ tends to the instantaneous rigidity $\mu^{(k)}$ as $|s| \rightarrow \infty$,

$$k_E(v) = k_2^{\text{elastic}}(\{\mu^{(k)}\}_{k=1}^\ell). \quad (23)$$

Thus k_E is computed from the elastic body obtained by replacing each complex shear modulus by its elastic rigidity. It depends on the rigidities but not on the viscosities, so

$$\frac{\partial k_E}{\partial \mu^{(k)}}(v_0) = \frac{\partial k_2^{\text{elastic}}}{\partial \mu^{(k)}} \Big|_{\mu^{(m)} = \mu_0^{(m)}, m=1, \dots, \ell}, \quad \frac{\partial k_E}{\partial \eta^{(k)}}(v_0) = 0. \quad (24)$$

The rigidity derivative on the right is evaluated symbolically or by finite differences on the elastic propagator solution at the baseline.

The fluid Love number k_f is not included in u , but it is determined by the spectral data through (2). Since only the solid rheologies are perturbed here and the geometry and density are held constant, k_f does not vary. The spectral derivatives must therefore satisfy

$$0 = \frac{\partial k_E}{\partial v_j} - \sum_{i=1}^{n+1} \left(\frac{1}{s_i} \frac{\partial r_i}{\partial v_j} - \frac{r_i}{s_i^2} \frac{\partial s_i}{\partial v_j} \right). \quad (25)$$

We use this identity to check the numerical implementation.

Collecting the derivatives gives the physical spectral Jacobian

$$J_v = Du(v_0) \in \mathbb{R}^{(2n+3) \times p}. \quad (26)$$

To make numerical diagnostics independent of physical units, we use relative perturbations,

$$\delta v_j = \frac{\Delta v_j}{v_{0,j}}, \quad \delta u_k = \frac{\Delta u_k}{u_{0,k}}, \quad (27)$$

whenever the corresponding baseline components are nonzero. The scaled Jacobian is

$$\widehat{J}_v = \text{diag}(u_0)^{-1} J_v \text{diag}(v_0). \quad (28)$$

Since the poles are negative, the relative pole perturbations are signed. Equivalently, $\Delta s_i/s_i$ may be read as $\Delta \log(-s_i)$ and $\Delta r_i/r_i$ as $\Delta \log r_i$, so the scaled sensitivities are logarithmic derivatives. All singular-value and resolution diagnostics below are defined using scaled variables unless stated otherwise.

4 DIFFERENTIATING THE EFFECTIVE HOMOGENEOUS REDUCTION

Section 3 gives the physical spectral Jacobian $J_v = Du(v_0)$. We now differentiate the algebraic inversion $u \mapsto w = \mathcal{I}(u)$ and compose the two differentials to obtain the local physical-to-effective map introduced in (15). Thus

$$\widehat{T} = \text{diag}(w_0)^{-1} D\mathcal{I}(u_0) J_v \text{diag}(v_0), \quad (29)$$

where the entries of $\widehat{T}$ measure relative changes of effective parameters induced by relative changes of physical layer rheologies. The inversion $u \mapsto w$ is algebraic, so its derivative can be computed directly from the pole-residue representation and the generalized Voigt inversion formulas.

4.1 Jacobian of the inversion

We compute the Jacobian of the pointwise inversion at the baseline spectral state u_0 by differentiating the pole-residue reconstruction and

the generalized Voigt formulas. For $u = (k_E, s_1, r_1, \dots, s_{n+1}, r_{n+1})$, write

$$P(s) = \prod_{i=1}^{n+1} (s - s_i), \quad N(s) = k_E P(s) + \sum_{i=1}^{n+1} r_i \prod_{j \neq i} (s - s_j), \quad (30)$$

so that these polynomials reconstruct the rational Love number. Using the normalization of (8), we regard

$$Q(s, u) = \frac{\Theta P(s) - \gamma(u) N(s)}{s}, \quad \gamma(u) = \frac{\Theta}{k_f}, \quad (31)$$

as a polynomial in s depending on the spectral coordinates u . The division by s is regular because the numerator vanishes at $s = 0$.

Using (4) and (2), the gravitational and elastic rows are

$$\begin{aligned} \frac{\partial \gamma}{\partial k_E} &= -\frac{\Theta}{k_f^2}, \quad \frac{\partial \gamma}{\partial r_i} = \frac{\Theta}{k_f^2 s_i}, \quad \frac{\partial \gamma}{\partial s_i} = -\frac{\Theta}{k_f^2} \frac{r_i}{s_i^2}, \\ \frac{\partial \alpha}{\partial u_k} &= -\frac{\Theta}{k_E^2} \frac{\partial k_E}{\partial u_k} - \frac{\partial \gamma}{\partial u_k}. \end{aligned} \quad (32)$$

Here $\partial k_E / \partial u_k$ equals one when $u_k = k_E$ and zero otherwise.

The isolated dashpot row follows from $\eta = Q(0)/N(0)$:

$$\frac{\partial \eta}{\partial u_k} = \frac{Q_{u_k}(0)N(0) - Q(0)N_{u_k}(0)}{N(0)^2}. \quad (33)$$

Here $Q_{u_k}(0)$ is obtained by differentiating the polynomial Q after the division in (8). Equivalently, $Q(0) = \Theta P'(0) - \gamma N'(0)$.

The simple roots q_i of Q define the Kelvin–Voigt elements:

$$Q(q_i(u), u) = 0. \quad (34)$$

Implicit differentiation gives

$$\frac{\partial q_i}{\partial u_k} = -\frac{Q_{u_k}(q_i)}{Q'(q_i)}, \quad (35)$$

$$Q_{u_k}(s) = \frac{\Theta P_{u_k}(s) - \gamma_{u_k} N(s) - \gamma N_{u_k}(s)}{s}, \quad s \neq 0.$$

Here $Q' = Q_s$, the partial derivatives P_{u_k} and N_{u_k} are obtained from (30), and γ_{u_k} is given by (32). The expression is evaluated only at the nonzero roots q_i .

Using

$$\eta_i = \frac{q_i Q'(q_i)}{N(q_i)}, \quad (36)$$

the chain rule gives

$$\begin{aligned} \frac{\partial \eta_i}{\partial u_k} &= \frac{q_i Q'_{u_k}(q_i) N(q_i) - q_i Q'(q_i) N_{u_k}(q_i)}{N(q_i)^2} \\ &\quad + \left[\frac{Q'(q_i) + q_i Q''(q_i)}{N(q_i)} - \frac{q_i Q'(q_i) N'(q_i)}{N(q_i)^2} \right] \frac{\partial q_i}{\partial u_k}, \end{aligned} \quad (37)$$

where $Q'_{u_k} = \partial_s Q_{u_k}$. Since $\alpha_i = -q_i \eta_i$, the spring row is

$$\frac{\partial \alpha_i}{\partial u_k} = -\frac{\partial q_i}{\partial u_k} \eta_i - q_i \frac{\partial \eta_i}{\partial u_k}. \quad (38)$$

Equations (32)–(38) assemble the rows of $D\mathcal{I}(u_0) = J_w^{-1}$ using all spectral and effective coordinates.

4.2 Composition with the physical tangent space

The Jacobian derived above is the differential of the full algebraic inversion with respect to the spectral coordinates. The physical perturbations considered in this paper occupy a smaller tangent space,

because the geometry and density are fixed and only the solid-layer rheologies are varied. In particular, the fluid Love number does not change. Therefore, for every physical perturbation Δv ,

$$Dk_f(v_0)\Delta v = 0. \quad (39)$$

Equivalently, the spectral Jacobian J_v satisfies the fixed- k_f identity (25).

Composing the algebraic inverse differential with the physical spectral Jacobian gives

$$T = D\mathcal{I}(u_0)J_v. \quad (40)$$

In relative variables this is the scaled map (29). Although $D\mathcal{I}(u_0)$ contains a nonzero γ row as a derivative with respect to arbitrary spectral perturbations, that row vanishes after composition with J_v . Indeed,

$$D\gamma(u_0)J_v\Delta v = -\frac{\Theta}{k_f^2}Dk_f(v_0)\Delta v = 0. \quad (41)$$

For rheological perturbations, the physical-to-effective map therefore has no gravitational component. Fixing the geometry and density imposes $Dk_f = 0$, so we work on the corresponding spectral tangent subspace.

In the implementation, the full Jacobian $D\mathcal{I}(u_0)$ is computed using the coordinates $(k_E, s_i, r_i) \mapsto (\gamma, \alpha, \eta, \alpha_i, \eta_i)$. The diagnostics below are evaluated after composition with J_v on the fixed- k_f physical tangent space. We check the implementation by verifying that the composed γ row is negligible, as required by (25), and by comparing $J_w^{-1}J_v$ with the symbolic composition $D\mathcal{I}(u_0)J_v$. Here $J_w = \partial u / \partial w$ is computed from the effective forward model.

4.3 Reduced and projected maps

Above, we derived the differential map for the complete algebraic reduction. In applications, the effective model may be reduced before it is used in an orbital or spin integration. We select the effective parameters at the baseline and keep them fixed during differentiation. This is the local form of the dominant-mode selection used in the pointwise reduction (Gevorgyan et al. 2023; Gevorgyan 2026c). Keeping the selection fixed avoids discontinuities when a perturbation moves an element across the dominance threshold.

Let w_{red} be the retained effective parameter vector. The target against which the reduced model is matched may be the reduced spectral state, the full spectral state, or a finite set of Love-number samples over a frequency interval. We represent this target by a differentiable map

$$\mathcal{G} : u \mapsto y, \quad (42)$$

where the real and imaginary parts are treated as separate components when y is complex. The corresponding physical and effective tangent maps are

$$J_{v,\mathcal{G}} = D\mathcal{G}(u_0)J_v, \quad J_{w,\mathcal{G}} = D\mathcal{G}(u_0)J_{w,\text{red}}. \quad (43)$$

Here $J_{w,\text{red}} = \partial u / \partial w_{\text{red}}$ is the spectral Jacobian with respect to the retained effective parameters, evaluated by freezing the omitted parameters at their baseline values.

If the reduced Jacobian is square and nonsingular for the chosen target, the reduced transfer matrix is

$$T_{\text{red}} = J_{w,\mathcal{G}}^{-1}J_{v,\mathcal{G}}. \quad (44)$$

More generally, when the retained model has fewer effective parameters than the target requires, we use the least-squares tangent

projection

$$\Delta w_{\text{red}} \approx J_{w,\mathcal{G}}^+ J_{v,\mathcal{G}} \Delta v, \quad (45)$$

where $J_{w,\mathcal{G}}^+$ is the Moore–Penrose pseudoinverse (Barata & Hussein 2012). The pseudoinverse gives the first-order change in the reduced effective parameters that minimizes the target mismatch in the chosen norm. This is a local least-squares projection for the selected reduced model.

4.4 Diagnostics

We interpret the local maps using singular values, resolution matrices, projection residuals, and mode-dominance weights. All diagnostics use relative physical perturbations. When the target is an effective-parameter vector, the effective parameters are scaled relatively. When the target is the complex Love number sampled over a frequency interval, we treat its real and imaginary parts as separate components without rescaling them individually. Although k_2 is dimensionless, its sampled real and imaginary parts can have very different scales.

Let A be any scaled tangent map whose columns correspond to the physical rheological parameters. For the full effective reduction, $A = \hat{T}$. For a reduced effective model, A is the corresponding scaled reduced map. For a direct frequency-interval diagnostic, A may instead be the scaled physical map from layer perturbations to Love-number samples. We write the singular value decomposition as

$$A = U\Sigma V^T, \quad (46)$$

and use the right singular vectors in V to form combinations of physical-layer perturbations and rank these directions by their corresponding singular values, in decreasing order of their first-order influence on the chosen target. When the target is an effective-parameter vector, the left singular vectors in U are combinations of effective rheological parameters. Retaining the leading right singular directions follows the dimension-reduction construction of Constantine et al. (2014), commonly called an active subspace.

To localize the physical directions retained by the map, we use the truncated resolution matrix

$$R_\tau = A_\tau^+ A_\tau = V_\tau V_\tau^T. \quad (47)$$

Here A_τ^+ is the pseudoinverse formed after discarding all singular values with $\sigma_i \leq \tau\sigma_1$, and V_τ collects the retained right singular vectors. The truncation is part of the diagnostic. If A has full column rank and no truncation is applied, then $A^+A = I$ and the resolution matrix carries no information. We therefore choose τ inside a clear gap in the singular spectrum. A diagonal entry of R_τ close to one means that the corresponding physical parameter lies in the retained subspace. A value close to zero means that the direction is nearly invisible.

Here R_τ measures model-to-model resolution: it determines which physical perturbations can be recovered from their changes to the chosen target. Observational resolution would additionally require a data operator and noise model (Backus & Gilbert 1970; Tarantola 2005).

For projected reduced models, we measure how much of the physical target sensitivity is not spanned by the retained effective tangent space. The projection residual is

$$\varepsilon_{\mathcal{G}} = \frac{\|\hat{J}_{v,\mathcal{G}} - \hat{J}_{w,\mathcal{G}}\hat{J}_{w,\mathcal{G}}^+\hat{J}_{v,\mathcal{G}}\|_F}{\|\hat{J}_{v,\mathcal{G}}\|_F}. \quad (48)$$

The hats indicate that scaling is applied before the pseudoinverse is formed. Scaling and pseudoinversion do not commute in general. The value $\varepsilon_{\mathcal{G}}$ measures the fraction of first-order target sensitivity that the retained effective model cannot reproduce.

When fitting the complex Love number over a frequency interval, we combine the Jacobians of its real and imaginary parts in a single least-squares problem. Let $\widehat{T}_{\mathcal{B}}$ be the resulting reduced transfer matrix. The corresponding complex residual matrix is

$$\mathcal{R}_{\mathcal{B}} = \widehat{J}_{v,\mathcal{B}} - \widehat{J}_{w,\mathcal{B}} \widehat{T}_{\mathcal{B}}. \quad (49)$$

We define

$$\varepsilon_{\mathcal{B}}^{\text{Re}} = \frac{\|\text{Re } \mathcal{R}_{\mathcal{B}}\|_F}{\|\text{Re } \widehat{J}_{v,\mathcal{B}}\|_F}, \quad \varepsilon_{\mathcal{B}}^{\text{Im}} = \frac{\|\text{Im } \mathcal{R}_{\mathcal{B}}\|_F}{\|\text{Im } \widehat{J}_{v,\mathcal{B}}\|_F}. \quad (50)$$

We evaluate both residuals using $\widehat{T}_{\mathcal{B}}$ from the combined least-squares problem.

We compute the mode-dominance matrix from the relative pole sensitivities in Section 3:

$$D_{ij} = \frac{\left| \frac{v_{0,j}}{s_{i,0}} \frac{\partial s_i}{\partial v_j} \right|}{\sum_{m=1}^p \left| \frac{v_{0,m}}{s_{i,0}} \frac{\partial s_i}{\partial v_m} \right|}. \quad (51)$$

For each mode, a row gives the relative contribution of every physical rheological parameter to its timescale sensitivity. If the denominator vanishes, we leave the row undefined or set it to zero in plots because none of the selected parameters changes that pole at first order.

5 A TWO-LAYER TOY MODEL

We use a two-layer body as a toy model.

5.1 The two-layer model

Consider an incompressible body with an effectively inviscid liquid core and a Maxwell viscoelastic mantle. The body and core radii, mass, and densities are fixed, so the only rheological parameters allowed to vary are those of the mantle,

$$v = (\eta_m, \mu_m)^T, \quad (52)$$

which enter the response through the mantle complex shear modulus

$$\widehat{\mu}_m(s) = \frac{\mu_m \eta_m s}{\mu_m + \eta_m s} = \frac{\mu_m s}{s + \omega_m}, \quad \omega_m = \frac{\mu_m}{\eta_m}. \quad (53)$$

The characteristic determinant is a quadratic polynomial in the mantle complex shear modulus $\widehat{\mu}_m$. After substituting (53), the characteristic equation can be written as

$$A(v) s^2 + B(v) s + C(v) = 0. \quad (54)$$

The corresponding Love number is

$$k_2(s) = \frac{N_2(v) s^2 + N_1(v) s + N_0(v)}{A(v) s^2 + B(v) s + C(v)}, \quad (55)$$

whose coefficients A, B, C and N_0, N_1, N_2 depend on η_m and μ_m through $\widehat{\mu}_m$ and $\omega_m = \mu_m/\eta_m$. The Love number has two relaxation poles, corresponding to a generalized Voigt model with one Voigt element.

5.2 Spectral sensitivities

For $v_j \in \{\eta_m, \mu_m\}$, applying (20) to the quadratic characteristic polynomial gives

$$\frac{\partial s_i}{\partial v_j} = -\frac{A_{v_j} s_i^2 + B_{v_j} s_i + C_{v_j}}{2A s_i + B}, \quad i \in \{+, -\}. \quad (56)$$

We compute the residues as

$$r_i = \frac{N_2 s_i^2 + N_1 s_i + N_0}{2A s_i + B} = \frac{N(s_i)}{P_s(s_i)}, \quad (57)$$

and their derivatives using (22). Together with k_E , these derivatives determine the two-layer physical spectral Jacobian J_v for the variables in (52).

5.3 Two-layer transfer matrix

The complete spectral and effective parameter vectors have equal dimension:

$$u_{2L} = (k_E, s_+, r_+, s_-, r_-)^T, \quad w_{2L} = (\gamma, \alpha, \eta, \alpha_1, \eta_1)^T. \quad (58)$$

The transfer matrix is

$$\Delta w_{2L} \approx T_{2L} \Delta v, \quad T_{2L} = D I(u_{2L,0}) J_v. \quad (59)$$

For the numerical toy model, an effectively inviscid liquid core with radius 325 km and density 6700 kg m⁻³ is surrounded by a homogeneous Maxwell mantle extending to the surface at $R = 1737.15$ km. We assign the mantle the upper-mantle values from Table 1: density 3356 kg m⁻³, rigidity $\mu_m = 62.5$ GPa, and viscosity $\eta_m = 10^{21}$ Pa s. The two poles correspond to relaxation times $\tau_i = -1/s_i$ of 3.0×10^4 yr and 3.4×10^5 yr. With the geometry and density fixed, k_f and γ do not vary. We evaluate the scaled transfer matrix numerically at this baseline. The γ row is below 2×10^{-8} and is omitted:

$$\widehat{T}_{2L} = \begin{matrix} & \eta_m & \mu_m \\ \begin{matrix} \alpha \\ \eta \\ \alpha_1 \\ \eta_1 \end{matrix} & \begin{pmatrix} 1 \times 10^{-4} & 1.0000 \\ 1.0000 & -4 \times 10^{-8} \\ -3 \times 10^{-4} & -1.6 \times 10^{-3} \\ 0.9997 & -3.1 \times 10^{-3} \end{pmatrix} \end{matrix}. \quad (60)$$

A relative change in mantle rigidity produces the same relative change in α . A relative change in mantle viscosity produces nearly the same relative changes in η and η_1 . All other matrix entries have magnitudes below 3.1×10^{-3} . The small sensitivity of α_1 indicates that it is determined mainly by the fixed geometry and density. In this model, α_1 measures the strength of the slow core–mantle buoyancy response, and η_1 controls its relaxation time.

6 FIVE-LAYER MODEL OF THE MOON

We use the same lunar model as in Gevorgyan et al. (2023) and Gevorgyan (2026c) to analyze the differential map of its effective homogeneous rheology.

We first use the full transfer matrix to determine which solid-layer rheological perturbations are visible in the complete Love-number response when expressed in effective-rheology coordinates. We then restrict the analysis to the frequency interval relevant to lunar laser ranging and determine which perturbations remain visible to the observations. Both analyses are local to the chosen baseline. We do not perform a new inversion of the lunar interior.

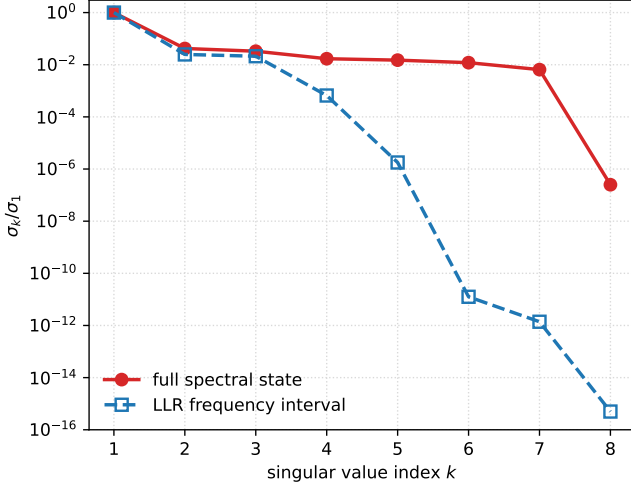


Figure 2. Singular-value spectra at the lunar baseline, normalized by the largest value. Filled circles show the full scaled transfer matrix $\hat{T}_{\text{full}}$ (62). Open squares show the frequency-interval physical map $\hat{J}_{\nu, \mathcal{B}}$ discussed in Section 6.4. Lines connect ordered singular values. The full map has a clear gap after the seventh singular value. Fewer directions remain visible in the LLR frequency interval.

6.1 Baseline model

The baseline consists of a solid inner core, a liquid outer core, a low-viscosity zone, an upper mantle, and a crust. We model the low-viscosity zone as a viscosity contrast at the base of the mantle rather than as the low-rigidity transition layer of Matsuyama et al. (2016) (Table 1).

We vary only the viscosities and rigidities of the solid viscoelastic layers:

$$\nu_{\text{full}} = (\eta_{\text{ic}}, \mu_{\text{ic}}, \eta_{\text{LVZ}}, \mu_{\text{LVZ}}, \eta_{\text{man}}, \mu_{\text{man}}, \eta_{\text{cr}}, \mu_{\text{cr}})^T. \quad (61)$$

The liquid outer core, radii, and densities are fixed. All diagnostics below use relative perturbations as in (27).

The lunar response has $n + 1 = 9$ relaxation poles, hence $n = 8$ Voigt elements plus the spring–dashpot pair (α, η) . We distinguish mode times $\tau_i = -1/s_i$ from the retardation times η_i/α_i .

6.2 Full map and resolution

For this baseline, we omit the fixed γ row and evaluate the scaled full transfer matrix

$$\hat{T}_{\text{full}} = \text{diag}(w_0)^{-1} D I(u_0) J_\nu \text{diag}(v_0). \quad (62)$$

We normalize its singular values by the largest one:

$$\sigma_k/\sigma_1 = \{1, 0.042, 0.033, 0.017, 0.015, 0.012, 0.0065, 2.5 \times 10^{-7}\}. \quad (63)$$

The first seven singular values remain within about two orders of magnitude. The eighth drops by more than four orders (Figure 2). We therefore find eight physical directions, seven response-sensitive and one nearly invisible.

The truncated resolution matrix,

$$R_{\text{full}} = \hat{T}_{\text{full}, \tau}^+ \hat{T}_{\text{full}} = V_7 V_7^T, \quad (64)$$

is formed by placing the threshold τ in the gap between σ_7/σ_1 and

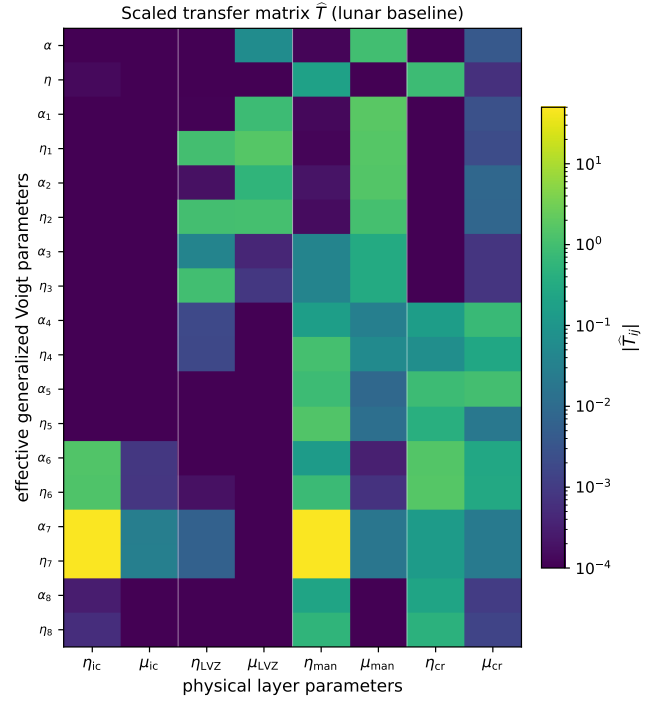


Figure 3. Magnitude of the scaled transfer matrix $\hat{T}_{\text{full}}$ at the lunar baseline. The color scale is logarithmic, and the vanishing γ row is omitted. Rows are effective generalized Voigt parameters and columns are the eight physical rheological parameters. The inner-core rigidity column μ_{ic} is uniformly near zero. The remaining columns contain non-negligible entries.

σ_8/σ_1 . Seven diagonal entries are equal to one within rounding error. The inner-core-rigidity entry is $R_{\mu_{\text{ic}}, \mu_{\text{ic}}} \approx 4 \times 10^{-7}$. The discarded right singular vector is aligned with the inner-core-rigidity axis to better than one part in 10^6 .

This alignment is consistent with the weak dependence of Love numbers on core elastic structure (Okubo & Saito 1983) and with lunar inversions in which degree-two tidal data do not resolve the inner core (Briaud et al. 2023). In the transfer-matrix heatmap (Figure 3), the μ_{ic} column is uniformly near zero. The other solid-layer parameters produce non-negligible patterns across the effective parameters. The inner-core viscosity remains visible because it controls a slow relaxation mode, as shown by the mode-dominance diagnostic in Section 6.3.

The large value $\sigma_1 = 91$ reflects a near-degenerate pair of Voigt elements with times near 1.2–1.3 Myr rather than an additional loss of rank. Along the corresponding singular direction, the inner-core and upper-mantle viscosities vary together and control the separation between the two relaxation times. When η_{ic} is increased from 10^{21} to 3×10^{21} Pa s, the largest singular value drops to 3.8. The seven-plus-one split remains unchanged under this and the other baseline-viscosity changes in Table 2. Changing the LVZ or crust viscosity shifts the corresponding relaxation times but does not change the rank structure of the full transfer map or the alignment of the near-null direction with μ_{ic} .

6.3 Mode dominance

Equation (51) compares the relative influence of the physical parameters on each relaxation pole. For a given pole s_i , the entries D_{ij} are

Table 1. Baseline parameters for the five-layer lunar Maxwell model. Radii, densities, and rigidities follow Matsuyama et al. (2016). Viscosities are taken from (Gevorgyan et al. 2023; Gevorgyan 2026c).

Layer	Outer radius km	Density kg m ⁻³	Rigidity GPa	Viscosity Pa s
Solid inner core	213.52	7720.16	40	10 ²¹
Liquid outer core	325.00	6700.00	10 ⁻¹⁰	1
Low-viscosity zone	500.00	3800.00	60	5 × 10 ¹⁶
Upper mantle	1697.15	3356.00	62.5	10 ²¹
Crust	1737.15	2735.00	15	10 ²³

Table 2. Robustness of the main full-map diagnostics under selected changes of the baseline viscosities. In all cases, the spectrum retains seven response-sensitive directions and one near-null direction aligned with the inner-core rigidity axis.

Variant	σ_8/σ_1	$R_{\mu_{ic}, \mu_{ic}}$	Main spectral change
Reference model	2.5×10^{-7}	3.8×10^{-7}	—
$\eta_{LVZ} \times 10$	2.2×10^{-7}	3.8×10^{-7}	LVZ timescales increase by 10
$\eta_{LVZ}/10$	2.5×10^{-7}	3.8×10^{-7}	LVZ timescales decrease by 10
$\eta_{cr} = 10^{21}$ Pa s	3.8×10^{-7}	3.8×10^{-7}	Crust-controlled modes move to mid-spectrum

Table 3. Relaxation times and dominant rheological controls of each relaxation mode at the lunar baseline, from the mode-dominance matrix (51). The entries are relative pole sensitivities, row-normalized, with contributions below 2% omitted. The mode times $\tau_i = -1/s_i$ are distinct from the Voigt retardation times η_i/α_i . Asterisks mark the two modes whose associated elements are retained by the frequency-interval model in Section 6.4.

Mode	τ_i	Dominant parameters (%)
s_1^*	12.2 d	η_{LVZ} 50, μ_{LVZ} 39, μ_{man} 11
s_2^*	18.3 d	η_{LVZ} 50, μ_{LVZ} 26, μ_{man} 24
s_3	63 yr	η_{LVZ} 99.8
s_4	1.9×10^4 yr	μ_{cr} 49, η_{man} 46, η_{cr} 4
s_5	2.3×10^4 yr	η_{man} 80, μ_{cr} 16
s_6	2.6×10^5 yr	η_{cr} 54, μ_{cr} 45
s_7	1.29×10^6 yr	η_{man} 70, η_{ic} 29
s_8	1.32×10^6 yr	η_{ic} 71, η_{man} 29
s_9	4.0×10^8 yr	η_{cr} 91, η_{man} 9

non-negative and sum to one over j . For example, $D_{ij} = 0.7$ means that v_j accounts for 70% of the total absolute first-order sensitivity of s_i among the selected parameters. At the lunar baseline, the modes separate into groups associated mainly with the low-viscosity zone, the upper mantle, the crust, and the inner-core viscosity (Table 3). We compute the matrix from pole sensitivities before applying the effective inversion, so the result concerns physical timescales rather than how they appear in generalized Voigt parameters.

Each row shows which physical parameters most strongly change a pole at first order. The absence of μ_{ic} is consistent with its near-zero resolution in Section 6.2. The presence of η_{ic} in s_8 explains why the inner-core viscosity remains visible in the full response.

6.4 Frequency-interval reduction

The complete pole-residue state represents the full effective rheology. For the Moon, we restrict the analysis to the frequency interval covered by lunar laser ranging (LLR) constraints, $\omega \in [3.3 \times 10^{-8}, 2.7 \times 10^{-6}]$ s⁻¹, corresponding to forcing periods from about one month to six years (Williams & Boggs 2015; Gevorgyan

et al. 2023). At frequencies ω_m across the interval, we evaluate

$$\mathcal{B}(u) = (k_2(i\omega_1), \dots, k_2(i\omega_M)), \quad (65)$$

$$k_2(i\omega) = k_E + \sum_{i=1}^{n+1} \frac{r_i}{i\omega - s_i},$$

where $M = 50$ frequencies are logarithmically spaced across the LLR frequency interval. The Jacobians contain separate rows for the real and imaginary parts. The derivative of the sampling operator has rows

$$\frac{\partial k_2(i\omega_m)}{\partial k_E} = 1, \quad (66)$$

$$\frac{\partial k_2(i\omega_m)}{\partial s_i} = \frac{r_i}{(i\omega_m - s_i)^2}, \quad (67)$$

$$\frac{\partial k_2(i\omega_m)}{\partial r_i} = \frac{1}{i\omega_m - s_i}. \quad (68)$$

The physical and effective frequency-interval maps are then

$$J_{v, \mathcal{B}} = D\mathcal{B}(u_0)J_v, \quad J_{w, \mathcal{B}} = D\mathcal{B}(u_0)J_w. \quad (69)$$

A reduced model contains a selected subset of the parameter pairs (α, η) and (α_i, η_i) . We retain the corresponding columns of $J_{w, \mathcal{B}}$ before forming the pseudoinverse in (48). We evaluate the projection residual $\varepsilon_{\mathcal{B}}$ from (48) with $\mathcal{G} = \mathcal{B}$. The number of required pairs depends strongly on the response being matched. Against the full spectral state, omitting any pair changes the sensitivity at the percent level. In the LLR frequency interval, the slow modes contribute negligibly to the first-order sensitivity. Keeping only the global spring-dashpot pair and the two elements associated with the starred modes s_1^* and s_2^* in Table 3 gives

$$\varepsilon_{\mathcal{B}} = 2 \times 10^{-7}. \quad (70)$$

The real-part Jacobian is 19.1 times larger than the imaginary-part Jacobian in the Frobenius norm. The componentwise residuals (50) are

$$\varepsilon_{\mathcal{B}}^{\text{Re}} = 1.7 \times 10^{-7}, \quad \varepsilon_{\mathcal{B}}^{\text{Im}} = 1.6 \times 10^{-6}. \quad (71)$$

The dissipative first-order sensitivity is reproduced to about two parts in 10⁶ over the interval.

The pair (α, η) must be retained because α carries the elastic part

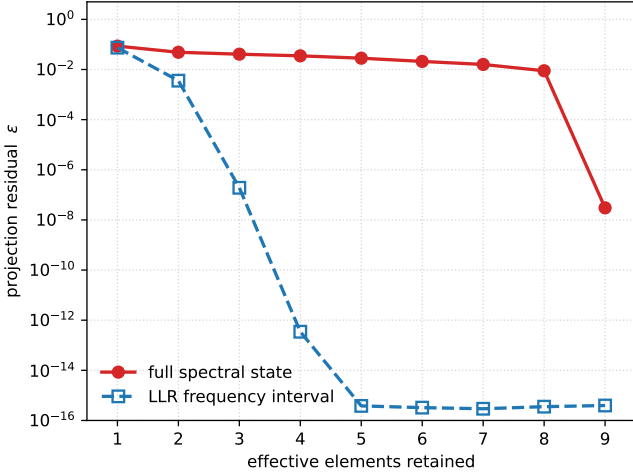


Figure 4. Projection residual (48) for nested reduced effective models obtained by successively discarding the least influential parameter pairs. The connected points give the resulting sequence of reduced models. The full spectral response retains sensitivity to all pairs. Over the LLR frequency interval, the pair (α, η) and the two elements associated with the starred modes listed in Table 3 reproduce the response.

of the response. Freezing this pair gives a frequency-interval residual of order 10^{-1} . The reduced frequency-interval model consists of this pair and the two elements associated with the 12.2 and 18.3 d Love-number modes (Figure 4).

The omitted modes describe slower relaxation processes, including the viscosity directions of the inner core and crust identified in Table 3. Their contributions to the Jacobian over the chosen interval are negligible. The singular spectrum of $\hat{J}_{v,B}$ in Figure 2 contains fewer visible physical directions than the full effective map.

We plot the resulting response as $Q(\omega)$ in Figure 5. The reduced model follows the full model through the LLR frequency interval and deviates below its lower bound, where the omitted slow modes begin to affect the response. We fit the complex Love number $k_2(i\omega)$ in (65).

6.5 Finite-perturbation check

We check the tangent prediction against finite perturbations of the physical parameters. For each coordinate direction e_j , we perturb the baseline as

$$v = v_0 + \delta v_{0,j} e_j, \quad (72)$$

rerun the full pointwise reduction, and compare the recomputed effective vector with the linear prediction

$$w_0 + T\Delta v. \quad (73)$$

To quantify the error, we compare relative changes in the complete effective parameter vector. For physical coordinate j , define

$$\delta w_j^{\text{num}} = \text{diag}(w_0)^{-1} [w(v_0 + \delta v_{0,j} e_j) - w_0], \quad (74)$$

$$\delta w_j^{\text{lin}} = \delta \hat{T} e_j,$$

We measure the relative error as

$$E_j(\delta) = \frac{\|\delta w_j^{\text{num}} - \delta w_j^{\text{lin}}\|_2}{\|\delta w_j^{\text{num}}\|_2}. \quad (75)$$

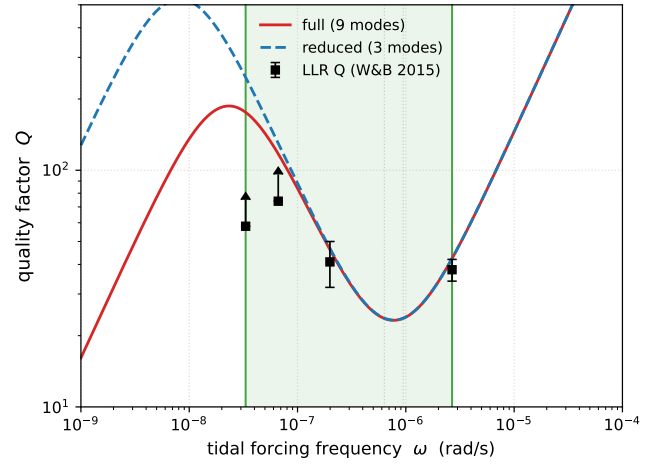


Figure 5. Lunar tidal quality factor $Q(\omega)$ for the full nine-mode effective model and for the reduced frequency-interval model retaining the pair (α, η) and the two elements whose modes lie within the interval. The fit uses the complex Love number $k_2(i\omega)$. The reduced model agrees with the full response through the LLR frequency interval and departs below its lower bound, where the omitted slow modes begin to contribute. Points are the lunar laser ranging determinations of Williams & Boggs (2015), compiled by Gevorgyan et al. (2023).

Because a perturbation can reorder the Voigt elements, we pair each recomputed element with the baseline element having the nearest retardation time η_i/α_i on a logarithmic scale. The errors quoted below are values of E_j . The maximum is taken over the eight physical parameters.

The prediction matches the recomputed effective parameters to about 2% or better for $\delta = 10^{-3}$. At $\delta = 10^{-2}$, six of the eight coordinate directions remain accurate to about 1%, but the inner-core and upper-mantle viscosity directions degrade to roughly 20%, because they control the nearly degenerate 1.2–1.3 Myr element pair discussed in Section 6.2. The tangent approximation remains within a few percent for $\delta \lesssim 10^{-3}$. For larger perturbations, errors first increase in these two viscosity directions, and the ordering of the two elements can change near $\delta = 10^{-1}$.

The fixed-geometry constraint is preserved throughout the finite-perturbation tests: the recomputed k_f and γ remain constant to better than 2×10^{-6} . A 10% perturbation of the inner-core rigidity changes the effective vector about thirty times less than the same perturbation applied to any other physical parameter.

7 DISCUSSION AND CONCLUSIONS

At the level of the frequency-dependent Love number, a finite multi-layer Maxwell body can be represented by a homogeneous body with generalized rheology (Gevorgyan et al. 2023). Here, we differentiate the mapping about a chosen layered baseline. The resulting local map quantifies how perturbations of the layer rheologies change the effective homogeneous parameters and which perturbations have little influence on the selected response.

We can use the transfer matrix for more than interpreting the effective parameters. Once an effective model has been constructed at v_0 , we estimate its response to prescribed small changes in the layer rheologies from $\Delta w \approx T\Delta v$. This local update avoids repeating the layered Love-number calculation and the algebraic inversion at every

step of a downstream orbital or rotational calculation. The update is valid only within the local range established by the finite-perturbation tests, and the baseline map must be recomputed when the evolving model leaves that range. If the uncertainties and correlations of the physical parameters are described by the covariance matrix Σ_v , then the corresponding effective-parameter covariance is, to first order,

$$\Sigma_w \approx T \Sigma_v T^\top. \quad (76)$$

We use the singular values and projection residuals to select a compact effective parameter set that retains the physical sensitivity over a chosen frequency interval.

Standard Love-number sensitivities directly describe how physical parameters perturb k_2 . We instead use the transfer matrix to express that local response change in the parameters of an effective rheology. A homogeneous time-domain solver can use the updated rheology and its uncertainty directly. The transfer matrix is not, by itself, an inversion of observations for interior parameters. Such an inference additionally requires an observational operator, data covariance, prior information, and regularization.

We distinguish quantities determined by the Love-number response from quantities that depend on the effective parameterization. The elastic limit, fluid limit, relaxation poles, and residues depend only on the Love-number response. If we change the effective parameterization, the numerical entries of the transfer matrix change. For example, generalized Voigt, Burgers, and finite Andrade models can describe the same local response using different parameter sets (Bland 2016; Gevorgyan 2021; Gevorgyan et al. 2023). The rank, physical near-null directions, and projection residuals therefore depend less on the chosen parameterization than the individual matrix entries.

For the complete effective representation, $D\mathcal{I}(u_0)$ is nonsingular on the rheological tangent space defined by fixed geometry and density. Consequently,

$$\ker T = \ker J_v, \quad (77)$$

and the homogeneous change of coordinates introduces no additional exact null directions. The near-null inner-core rigidity direction found in the full map reflects the weak sensitivity of the Love-number response itself. Additional response information is lost when we analyze only part of the response or omit effective elements. Truncating small singular directions affects only the diagnostic resolution adopted for interpretation.

In the lunar model, the full transfer map resolves seven of the eight solid-layer rheological directions and isolates the inner-core rigidity as the single near-null direction. This result agrees, in effective-rheology coordinates, with the weak sensitivity of degree-two Love numbers to the elastic structure of the inner core (Okubo & Saito 1983; Briaud et al. 2023). We obtain the same seven-plus-one split after the selected changes to the baseline viscosities.

We use the mode-dominance diagnostic to determine which physical rheological parameters most strongly move each pole at first order, without assigning each effective mode to a single geological layer. In the lunar model, the low-viscosity zone, mantle, crust, and inner-core viscosity affect distinct relaxation times. Over the frequency interval constrained by lunar laser ranging, the first-order response is reproduced by the pair (α, η) and the two Voigt elements associated with the modes within the interval. Contributions from the slower modes are negligible over this interval.

The equivalence considered here is an equivalence of the external linear response, not of the internal deformation fields. Matching the complex Love number reproduces the surface response and the total degree-two tidal dissipation for the same forcing, but it does not

reproduce the radial distribution of stress and strain. Consequently, the layer-wise or local dissipation cannot be recovered uniquely from the effective homogeneous parameters alone. Different layered interiors can produce nearly identical Love numbers but partition the dissipated energy differently among their layers.

This distinction between global and local dissipation is well established. Tobie et al. (2005) compute the global dissipated power from $\text{Im } k_2$ and determine the radial distribution of dissipation using sensitivity functions derived from the radial deformation functions. Beuthe (2013) represents spatial heating in a spherically stratified body using radial weight functions that depend on the internal structure and angular patterns determined by the tidal forcing. Integrating the corresponding local dissipation density over each layer gives the layer fractions used below.

In a coupled orbital–thermal model, we could supplement the effective rheology with dissipation-partition information calculated from the layered baseline using such radial sensitivity or weight functions. If $f_\ell(\omega; v)$ is the fraction of the total dissipation deposited in layer ℓ , the simplest local approximation is

$$\dot{E}_\ell(\omega; v) \approx f_\ell(\omega; v_0) \dot{E}_{\text{tot}}^{\text{eff}}(\omega; v), \quad (78)$$

where the effective model supplies the total dissipation. For a first-order update, we may use

$$f_\ell(\omega; v_0 + \Delta v) \approx f_\ell(\omega; v_0) + D_v f_\ell(\omega; v_0) \Delta v. \quad (79)$$

The heating prescription remains local to the selected baseline and requires additional layered-model calculations. The partition functions are not contained in the Love number or in the transfer matrix developed here.

The retained poles must be simple and remain separated under perturbation. As poles approach degeneracy, individual pole derivatives become ill-conditioned, and one should work with invariant subspaces rather than individual modes (Nelson 1976). At the lunar baseline, the nearly degenerate 1.2–1.3 Myr element pair limits the range of the tangent approximation. We assume a rational response. This is exact for finite multilayer Maxwell bodies (Sabadini et al. 2016; Gevorgyan et al. 2023). Rheologies with continuous relaxation spectra, such as Andrade or Sundberg–Cooper models, give non-rational Love numbers. The finite-dimensional differential method developed here can be applied after representing the response by a finite generalized Voigt model over the frequency interval of interest (Gevorgyan et al. 2020; Gevorgyan & Ragazzo 2026).

An observation-limited extension would combine the Jacobians derived here with an observation operator to determine which layered and homogeneous response directions can be distinguished from finite data, such as measurements of k_2 , Q , k_2/Q , moment of inertia, or libration amplitude.

Effective homogeneous parameters are therefore meaningful only relative to a baseline, a parameterization, and the response quantity being considered.

ACKNOWLEDGEMENTS

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DATA AVAILABILITY

No new observational data were generated for this study. The code used to compute the spectral sensitivities, transfer matrices, diagnostics, and figures is available at <https://github.com/>

ygevorgyan/local-differential-map and archived on Zenodo with a citable DOI (Gevorgyan 2026b). The homogeneous-rheology construction used as the pointwise reduction workflow is archived separately (Gevorgyan 2026a).

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